Kepler’s laws

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use orbital shape, equal areas and period comparisons.

SUCCESS CRITERIA - I can...

1. Place the central body at a focus of an ellipse.
2. Compare speed near and far from the central body.
3. Use $T^2=a^3$ only with the specified units and central body.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “period” and “ellipse” to explain your answer. Sentence starters are optional.

period: time for one complete cycle or orbit

ellipse: an oval-shaped closed curve with two focal points

orbit: a path around another body under gravity

01 Warm-up: bring back an earlier idea

A launch has v_x=**5 m/s** and v_y, i=**30 m/s** upward.
1. Use g=**10 m/s²**.
2. **Find time** to the top and total time back to launch height.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

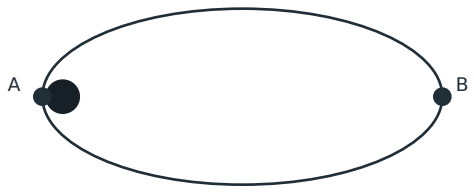
02 Worked example: explain the prediction

Reference: $T^2 = a^3$.
For Sun orbits only when T is in years and a is in AU. a is the semi-major axis. Equal areas are swept in equal times.

Worked reasoning: For an ideal Sun orbit, a=**4 AU** gives T=√(**4³**)=**8 years**. The orbit period depends on semimajor axis, not on a chosen point's instantaneous speed alone.

Your turn: Draw an ellipse, put the Sun at a focus and **mark the faster** and slower parts.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .

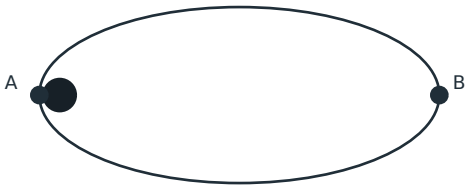


Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Mark the Sun at a focus on an ellipse. **Label where the object** is fastest and slowest.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .



Marzano check - circle: ① ② ③

04 Apply the physics

Supplied comparisons: a=**1 AU** gives T=**1 year**; a=**4 AU** gives T=**8 years**. Which orbit takes longer? **How many** Earth years for two orbits of the outer object?

Sentence starter: I notice ... . This supports ... because ... .

Semimajor axis (AU)	Period (years)
1	1
4	8

Marzano check - circle: ① ② ③

Kepler’s laws

antocciphysics

Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

Does knowing an orbit’s period alone tell the date when an asteroid reaches Earth’s location? **What else** is needed?

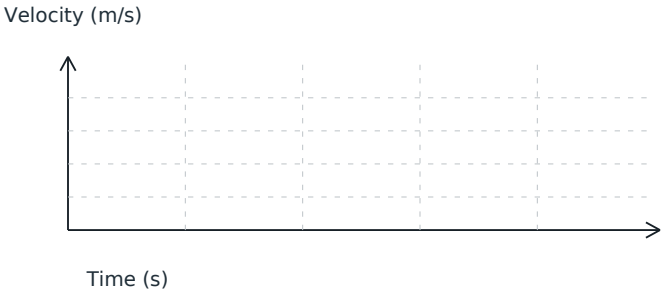
Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A ball is dropped from rest and falls ideally for **5 s**. Use g=**10 m/s²**. **Find its speed** and direction.
Sketch the motion and **label a positive direction**.

Sentence starter: I found … [unit]. This value means … in this situation.

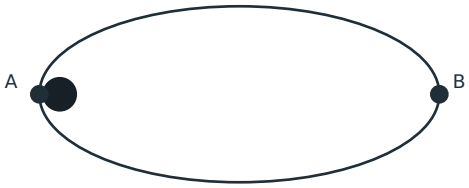


Marzano check - circle: 1 2 3

07 Exit check

An asteroid is nearest the Sun at point A and farthest at B. At which point is it fastest? **Name which law helps explain** this.

Sentence starter: I think … . I can point to …, which shows …, so … .

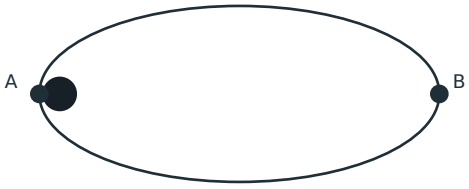


Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Two Sun orbits have supplied periods **1** and **8 years**. Their semimajor axes are **1** and **4 AU**. **Explain why four times the** size does not mean four times the period.

Sentence starter: I think … . I can point to …, which shows …, so … .



Marzano check - circle: 1 2 3